

SOFTWARE DESIGN DOCUMENT

DSP Project Template (Web-Based Application)

Project Title

AI-Based Deepfake Interaction System for Entertainment

Course

CENG 384 - Intro. To Signal Processing

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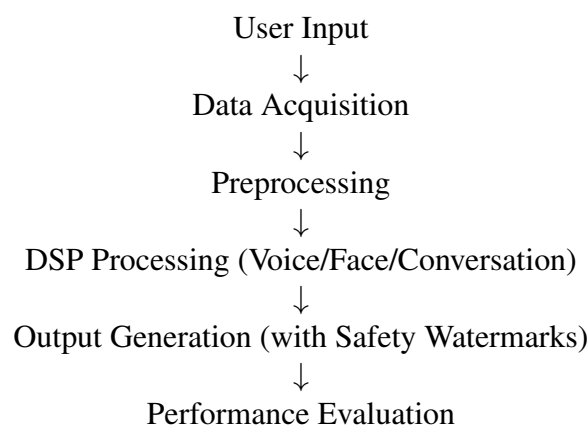
1 Introduction

1.1 Purpose

This Software Design Document outlines the underlying architecture, module configurations, and technical design decisions for the DSP-based web application developed for this project. Specifically, it details the web-based AI deepfake interaction platform designed for entertainment and academic demonstration, which encompasses text-to-speech voice cloning, video face swapping, and quantitative media evaluation.

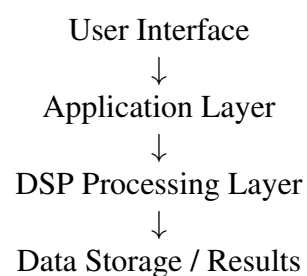
2 System Overview

System Workflow



3 System Architecture

Platform independent project architecture

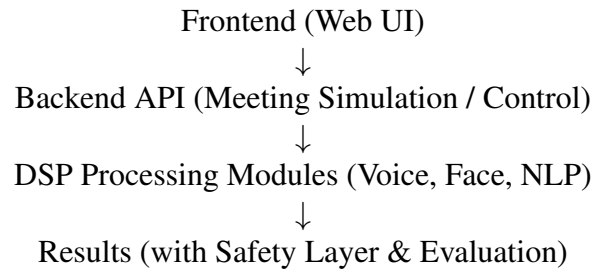


3.1 Platform Options

Platform	Example Technologies
Web	HTML, CSS, JavaScript, Python backend (Flask/FastAPI)

3.2 Architecture Models

Web Architecture



4 Module Design

4.1 Input Module (User Interface Layer)

Responsibilities:

- Receive user input via web interface (text prompts, microphone audio, webcam video).
- Load predefined celebrity models.
- Validate input format and report unsupported files.

Inputs: Audio (WAV/MP3), Image/Video (MP4/JPG), Text Streams.

Outputs: Validated raw data arrays ready for server-side processing.

4.2 Preprocessing Module

Functions:

- Face detection and bounding box alignment.
- Audio noise reduction and vocal normalization.
- Resampling video frames for real-time latency (under 500ms).

4.3 DSP Processing Module

Possible algorithms:

Category	Example Algorithms
Signal/Audio Analysis	Text-to-Speech (TTS), Mel-spectrogram inversion, Voice Conversion (VC).
Image/Video Processing	Generative Adversarial Networks (GANs), Facial landmark extraction.
NLP/Conversation	Large Language Models (LLMs) for personality simulation.

4.4 Output Module (Watermark & Safety Control Layer)

Functions:

- Generate processed deepfake results for web display.
- Embed inaudible audio watermarks and permanent visual watermarks (e.g., "Parody Mode").
- Restrict direct public auto-sharing.

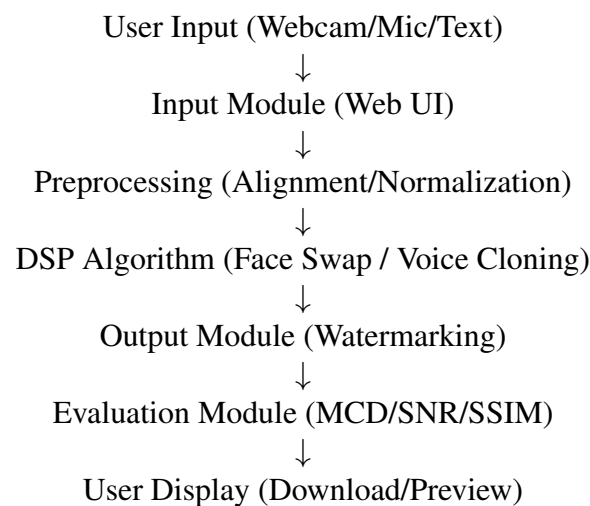
Outputs: Processed MP4/WAV files, Graphical UI previews.

4.5 Evaluation Module

Performance evaluations. Metrics (example):

Metric	Purpose
MCD (Mel-Cepstral Distortion)	Voice output error measurement.
SNR (Signal-to-Noise Ratio)	Generated audio quality analysis.
SSIM (Structural Similarity Index)	Video structural similarity evaluation.
PSNR (Peak Signal-to-Noise Ratio)	Reconstructed video quality evaluation.
Latency	End-to-end processing time (target: < 2 seconds for preview).

5 Data Flow Diagram



6 Algorithms

Example: Voice Cloning (DSP/Audio Processing)

Steps:

1. Extract linguistic features from input text or phonetic features from input audio.
2. Map features to the acoustic profile of the target celebrity model.
3. Synthesize waveform output using a vocoder.

Example: AI Video Face Swap

Steps:

1. Detect and extract facial landmarks from source video.
2. Map source expressions and head pose to the target identity latent space.
3. Blend the generated target face seamlessly back into the original frame lighting.

7 Technologies and Tools

Tool	Purpose
Python (Flask/FastAPI)	Core backend server and DSP processing.
OpenCV / Dlib	Image processing, face detection, and alignment.
PyTorch / TensorFlow	Running GANs and deep learning models.
Librosa	Audio processing, resampling, and spectrogram analysis.
Matplotlib / Web Charting	Visualization of evaluation metrics (SNR, MCD).

8 User Interface Design

Possible features:

- Web-based file upload and live webcam/microphone toggle.
- Celebrity selection panel (Voice, Face, Conversation style).
- Parameter adjustment (Speaking rate, pitch, emotional tone).
- Real-time preview with mandatory synthetic warning labels.
- Output display and secure download controls.

9 Testing and Evaluation

Test Case	Description
Input validation	Check supported formats (WAV, MP3, MP4) and corrupt file handling.
Algorithm test	Verify MCD, SNR, and SSIM scores meet minimum academic thresholds.
Performance test	Measure processing time to ensure live simulation latency is $< 500\text{ms}$.
Reliability test	Ensure stable processing for continuous sessions without crashing under server RAM limits.
Safety test	Verify watermarks cannot be bypassed during export from the web interface.

10 Expected Results

Examples:

- Realistic deepfake video and audio outputs restricted to parody/educational use.
- A fully functional, browser-based interface that successfully bridges simple user inputs with complex backend DSP and deep learning algorithms
- Graphical analysis reports detailing SSIM, PSNR, MCD, and SNR scores.

11 Limitations

Possible limitations:

- **Processing time:** High computational cost for real-time video rendering on the backend.
- **Server constraints:** Dependent on hosting hardware; concurrent users may experience lag.
- **Data availability:** Limited to predefined, publicly available celebrity datasets.
- **Ethical constraints:** Strict mandatory watermarking and prohibition of undisclosed impersonation.